



Genome-scale models

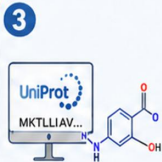
eciML1515 (*E. coli*)
Yeast9 (*S. cerevisiae*)



Candidate generation

16,613 model-linked
enzyme–reaction–
metabolite associations

8,797 *E. coli* | 7,816 yeast



Protein and molecular mapping

UniProt single-protein
sequences

selected reaction-
participant SMILES



Experimental k_{cat} matching

BRENDA + SABIO-RK

1,354 matched
associations

hierarchical species /
EC / UniProt /
substrate matching



Final benchmark filtering

- positive k_{cat} convertible to s^{-1}
- single valid protein sequence
- RDKit-parseable SMILES



Final benchmark and unified evaluation

1,246 benchmark records

781 *E. coli* | 465 *S. cerevisiae*

12 methods | 6
inference regimes